

Recall from linear algebra Given a matrix $A \in \mathbb{R}^{n \times n}$
(or a linear operator $A: \mathbb{R}^n \rightarrow \mathbb{R}^n$), the operator norm
(or induced norm) induced by the 2-norm $\|\cdot\|$ on $\mathbb{R}^n$) is

$$\begin{aligned} \|A\| &:= \sup_{\|x\|=1} \|Ax\| = \sup_{\|x\| \leq 1} \|Ax\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} \\ &\in [0, \infty) \\ &= \max \dots \end{aligned}$$

Exercise the equalities above are true.

Fact $\|Ax\| \leq \|A\| \|x\| \quad \forall x \in \mathbb{R}^n.$

 You might find this useful for HW 4 #1(c)
and for understanding the pf of $[C' \Rightarrow \text{loc. L.p.}]$ from Lec 15.

For rest of today: $E \subset \mathbb{R}^n$ is open, $f: E \rightarrow \mathbb{R}^n$ is loc. Lip., $I_x \subset \mathbb{R}$ is the maximal interval for any $x \in E$ (w.r.t. $\dot{y} = f(y)$, $y(0) = x$),

$$\Omega = \bigcup_{x \in E} I_x \times \{x\} = \{(t, x) \mid x \in E, t \in I_x\} \subset \mathbb{R} \times E,$$

and $\phi: \Omega \rightarrow E$ is the flow of f .

Def We will call ϕ a global flow if $\Omega = \mathbb{R} \times E$ (or, equivalently, if $I_x = \mathbb{R} \ \forall x \in E$).

Prop

(Properties of I_x ;
"group property" of
 Φ)

Let $x \in E$ and $t \in I_x$. Then

$$I_{\Phi^t(x)} = I_x - t := \{r - t \mid r \in I_x\},$$

so that $s + t \in I_x \iff s \in I_{\Phi^t(x)}$.

Moreover, if $s + t \in I_x$, then

$$\Phi^s(\Phi^t(x)) = \Phi^{s+t}(x) \quad \text{"group property"}$$

pf

Define $\alpha: I_{\Phi^t(x)} \longrightarrow E$ and $\beta: (I_x - t) \longrightarrow E$

by $\alpha(s) := \Phi^s(\Phi^t(x))$ and $\beta(s) := \Phi^{s+t}(x)$.

Note that $\alpha(0) = \Phi^t(x)$, $\beta(0) = \Phi^t(x)$, and

$$\dot{\alpha}(s) = f(\Phi^s(\Phi^t(x))) = f(\alpha(s)), \quad \dot{\beta}(s) = f(\Phi^{s+t}(x)) = f(\beta(s)).$$

Thus, $\alpha : I_{\phi^t(x)} \rightarrow E$ and $\beta : (I_x - t) \rightarrow E$ are both solutions to the IVP " $\frac{dy(s)}{ds} = f(y(s))$, $y(0) = \phi^t(x)$ ",

so the uniqueness theorem $\Rightarrow$ α and β coincide on the intersection of their domains.

Maximality of $I_{\phi^t(x)} \Rightarrow I_x - t \subset I_{\phi^t(x)}$.

Since $0 \in I_x$, this implies that $-t \in I_{\phi^t(x)}$,

so $x = \phi^0(x) = \beta(-t) = \alpha(-t) = \phi^{-t}(\underbrace{\phi^t(x)}_z)$.

Thus, defining $z := \phi^t(x)$ and repeating the above argument with (t, x) replaced by $(-t, z)$ yields

$$\underbrace{I_z}_{\phi^t(x)} - (-t) \subset I_{\underbrace{\phi^{-t}(z)}_x} = I_x, \text{ or}$$

equivalently

$$I_{\phi^t(x)} \subset I_x - t.$$

Combining this w/ the above yields

$$I_{\phi^t(x)} = I_x - t.$$



Thus, the domains of α and β are the same,
so $\alpha = \beta$. Since $\phi^s(\phi^t(x)) = \alpha(s) = \beta(s) = \phi^{s+t}(x)$,
we are done.



Lem If $K \subset E$ is cpet, then $\exists \varepsilon > 0$ s.t.
 $(-\varepsilon, \varepsilon) \times K \subset \Omega$. (In other words, $(-\varepsilon, \varepsilon) \subset I_x$ for all $x \in K$.)

pf For each $x \in K$, Picard-Lindelöf +
furnishes $\varepsilon_x, \delta_x > 0$ s.t. $(-\varepsilon_x, \varepsilon_x) \times B_{\delta_x}(x) \subset \Omega$.
We are crucially using the "+" portion here.

Note that $\{B_{\delta_x}(x)\}_{x \in K}$ is an open
cover of K . Since K is cpet, $\exists$ a finite subcover,
i.e., $\exists x_1, \dots, x_m \in K$ s.t. $K \subset \bigcup_{i=1}^m B_{\delta_{x_i}}(x_i)$.

Define $\varepsilon := \min_{1 \leq i \leq m} \varepsilon_{x_i} > 0$. Then since

$$(-\varepsilon, \varepsilon) \times B_{\delta_{x_i}}(x_i) \subset (-\varepsilon_{x_i}, \varepsilon_{x_i}) \times B_{\delta_{x_i}}(x_i) \subset \Omega \quad \forall i,$$

We have

$$(-\varepsilon, \varepsilon) \times K \subset (-\varepsilon, \varepsilon) \times \bigcup_{i=1}^m B_{\delta_{x_i}}(x_i) = \bigcup_{i=1}^m (-\varepsilon, \varepsilon) \times B_{\delta_{x_i}}(x_i) \\ \subset \Omega. \quad \square$$

Def A map $F: M \rightarrow N$ between metric spaces
is proper if $F^{-1}(K) \subset M$ is compact
for any compact subset $K \subset N$.

Recall
 $F^{-1}(K) := \{m \in M : F(m) \in K\}.$

Theorem
 (Either $I_x \cap [0, \infty) = [0, \infty)$, or
 the map $t \mapsto \phi^t(x)$,
 $I_x \cap [0, \infty) \rightarrow E$
 is proper)

Let $x \in E$ & $a, b \in [0, \infty) \cup \{+\infty\}$ be s.t.
 $I_x = (-a, b)$. Either $b = +\infty$ or
 the map $[0, b) \rightarrow E$ defined by
 $t \mapsto \phi^t(x)$ is proper.

pf We need to show that, if $b < \infty$ and
 $K \subset E$ is compact, then the set of
 $t \in [0, b)$ w/ $\phi^t(x) \in K$ is cpct, i.e., it has
 an upper bound strictly less than b .

By the Lem above, $\exists \varepsilon > 0$ s.t. $(-\varepsilon, \varepsilon) \subset I_z \quad \forall z \in K$.
 Suppose $\exists t > b - \varepsilon$ with $\phi^t(x) \in K$. Then $(-\varepsilon, \varepsilon) \subset I_{\phi^t(x)} = I_x - t$
 by the above Prop, so $(t - \varepsilon, t + \varepsilon) \subset I_x = (-a, b)$, contradicting
 the fact that $t + \varepsilon > b$. Thus, $b - \varepsilon$ is the desired upper bound. $\square$

Remark there is also a "backward time" version of above thm, which can be proved either by repeating the above proof (mutatis mutandis) or considering the "time-reversed" flow $\psi^t = \phi^{-t}$ of $-f$.